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Novel metamaterial platform with piezoelectric sensors for self-sensing mechanical support

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ABSTRACT

In the last two decades, the field of mechanical engineering has seen growing interest in the development and utilisation of mechanical metamaterials. Artificially designed materials are intended to alter the mechanical properties of structures significantly, thereby enhancing their resilience, adaptability, and efficiency. This research focuses not only on strengthening structural strength and durability but also on enabling structural health monitoring and vibration mitigation. Modern computational modelling, with coupled-field analysis, allows engineers to design smart systems that integrate piezoelectric elements into complex geometric structures. These smart piezoelectric elements, here embedded in auxetic reentrant unit cells, offer valuable insights into the behaviour of the host structure under various conditions. This integration facilitates the assessment of electromechanical responses, thereby enabling the development of more intelligent and responsive structural systems. The current research focuses on developing a computational model and creating experimental prototypes for self-sensing mechanical support. This support serves as a load-bearing element of the host structure while simultaneously enabling the generation of vibrational signals in response to external stimuli. Piezoceramic elements enable the support to function as a sensor, detecting external forces or environmental vibrations. Additionally, this structure opens new possibilities for studying mechanical vibration attenuation and the temporal decay of vibrations. The combination of advanced metamaterial design, computational tools, and integrated smart materials creates a new approach for structural health monitoring and vibration attenuation. Ultimately, such a system aims to develop a better understanding of sustainable structures that can adapt to and respond to their environment while maintaining optimal structural rigidity and functionality.

1. Introduction

Mechanical metamaterials offer a different perspective on structural design and an emphasis on altering mechanical properties, such as a negative Poisson's ratio, low density, and energy absorption. These unique properties have sparked interest in various engineering fields [1–3]. Auxetic structures have gained significant attention across scientific fields, including medicine and engineering. To further advance active mechanical metamaterials, a study of smart metamaterials offers a promising opportunity to integrate artificially designed structures with structural materials that possess additional physical-domain characteristics. The integration of smart materials, such as piezoelectric or thermal materials, offers an opportunity to create advanced smart structures or adaptive systems. These systems integrate advanced sensor technologies and actuators to support controllability, external-stimulus sensing, and structural health sensing within a single system over its lifetime, thereby improving overall performance.

The study of smart structures and materials has gained significant attention, and researchers have investigated these areas to meet the definition of smart structures: non-biological, tangible structures that have a specific purpose, utilise resources to achieve that purpose, are inspired by biological systems, and adapt optimally to changing environmental conditions. Spillman [4] played an early role in defining smart structures, and compared various terms for this concept, concluding that smart materials constitute a subset of smart structures.

Smart structures constitute an emerging field in engineering, encompassing multiple categories that intersect in a Venn diagram illustrating the relationship between adaptive and sensory structures. The Venn diagram describes well the components and intersections of various levels of smart structures, leading to the goal of intelligent

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structures [5]. The concept of smart structures intersects two fields: sensory and adaptive structures. The intersection represents controlled structures. Controlled structures comprise a subset of active structures and, within that subset, an even smaller subset of intelligent structures. Adaptive structures are characterised by the direct use of control elements to influence structural properties, whereas sensory structures comprise networks of sensors for structural monitoring. The Venn diagram shows that smart structures occupy both regions, leveraging the synergy between control elements and sensors to maximise functionality and enable effective responses to changes in the surrounding conditions. This integration yields controlled structures with feedback systems that permit active adjustment of structural characteristics based on sensor readings, thereby creating a category of controlled active structures that also includes load-bearing structures. Finally, intelligent or smart structures represent a more sophisticated concept, incorporating advanced control logic and electronics. By recognising these interconnected categories within smart structures, engineers can design and implement systems that meet specific requirements while benefiting from cost savings, improved performance, and reduced weight [6].

2. State of the art

Piezoelectricity plays a significant role in the functionality of these structures, particularly in sensors and actuators. Piezoelectric materials are among the smart materials that offer sensing capabilities and the ability to control material characteristics owing to the piezoelectric effect. This effect enables mechanical forces to create an electrical potential proportional to the polarisation [6–8]. Smart structures with piezoelectric elements leverage this property for various applications, such as monitoring mechanical events or converting electrical energy into mechanical energy in the form of displacement or transformation that affects the surrounding structures [9]. Key parameters for smart sensors and actuators include sensitivity, operational range, linearity, temperature sensitivity, brittleness, fatigue resistance, repeatability, reliability, power density, compactness, heat generation, sensitivity to various energy fields, and efficacy [10,11]. The motivations for research in these areas include potential reductions in production costs, weight reduction, precise information on structural health, vibration damping [12–15], and the prevention of structural failure. Smart materials have wide applications because of their unique physical properties that provide a respond to various external stimuli or behave advantageously to environmental changes, such as structural rigidity, multidomain dependency, enhanced performance, and energy savings compared to traditional materials.

Structural Health Monitoring (SHM) techniques are categorised into two main groups: destructive and nondestructive [16]. Current research mainly focuses on nondestructive testing, leading to various areas of interest such as data acquisition and signal processing [17–19], algorithms for predictive maintenance [20–22], health monitoring [23–25], sensor placement methods [26], sensor types [27–29], sensor embedding [30–34], and potentially self-healing properties [35,35]. SHM is relevant across different engineering fields, including aerospace [36–39], civil engineering [27,40–42], mechanical engineering [13,38,43], and medicine [44,45].

To achieve the predictive maintenance of structures, a network of sensors is usually deployed via external attachment, involving a network of wires and computational nodes [43,46,47]. Predictive maintenance relies on periodic or continuous monitoring of the system of interest [36]. Environmental factors affect the reliability of external sensors. Therefore, embedding sensors within load-bearing and functional structures to address the ecological effects is of utmost importance [48]. These structures require environment-resistant sensors that eliminate the need for wire networks, focus on load-bearing capabilities, and efficiently integrate sensors inside the structures [49,50]. Smart structures and materials, or adaptronics, focus on these topics [51,52].

Several engineering disciplines have delved into the research and development of smart structures and the integration of smart materials into stimuli-responsive systems. These systems aim to modify a structure's mechanical behaviour while preserving its original functionality and providing information on its current functional state, typically using computational modelling. Civil engineering focuses on smart concrete with integrated piezoelectric materials [53], concrete composites with carbon nanotubes (CNTs) [54], and smart concrete blocks for implementation in civil structures [55]. Aerospace and mechanical engineering studies the capability of smart composite structures with embedded smart materials, such as fibre Bragg grating (FBG) sensors [56], flexible piezoelectric materials [57], and CNTs [58]. Medical science is interested in the creation of multifunctional materials and designs for bone replacement or stimuli-responsive materials [59,60].

In all of these fields, various other variables also influence the design of smart structures. Self-healing materials are of interest in civil engineering [61]. Aerospace, mechanical engineering, and medical sciences seek superior mechanical properties in materials or design structures to achieve lightweight [62] or tunable mechanical properties [63–66]. The focus on enhanced mechanical properties and SHM principles may lead to the use of mechanical metamaterials [67,68] and smart materials [69–71] to achieve smart structures with superior capabilities [33,72]. The integration of smart and mechanical metamaterials offers an opportunity to develop novel structures that address various engineering challenges and meet the need for SHM [73]. In recent years, there has been substantial progress in the field of smart structures. Wang et al. [74] investigated a piezoelectric stack energy harvester based on an auxetic reentrant geometry that prevents buckling. They have sought to address the limitations of stand-alone piezoelectric stacks and other hinge-dependent transducers. They designed a force transducer that converts a compressive loading axis into a perpendicular axis. This increases the loading force, which depends on the truss's convex angle, and enhances the power output of the piezoelectric stack by up to 112-fold relative to the stand-alone configuration. This force amplifier enhances sensitivity, such that even a small load produces a larger voltage suitable for sensing and energy harvesting. The integration of piezoelectric material within a mechanical metamaterial geometry can highly enhance the sensing capabilities of a smart structure. Xu et al. [75] employed a cymbal mechanical metamaterial, integrated a PMNT piezoelectric single crystal within the cymbals, and created a smart cantilever that amplifies base-excited vibrations with various proof masses. They have enhanced the voltage generation of cymbal geometry and effectively employed force amplification. The advantage of integrating piezoelectrics within a structure with a substantial proof mass is that it can effectively lower the operating frequency and enable the use of high-frequency piezoelectric materials at lower frequencies. Another advantage of piezo-mechanical coupling is the ability to control the mechanical behaviour of the structure through the piezoelectric material. The use of PZT elements can enable active or passive vibration control of smart structures and mitigate critical vibrations [13].

In this paper, we designed a lightweight, auxetic metamaterial platform that embeds piezoelectric ceramic (PZT) transducers to provide simultaneous load-bearing support and self-sensing capabilities. Computational modelling provided an important validation tool in the combination of PZT elements and additively manufactured lattices from Ti-6Al-4 V and SS316L. Not only stiffening the lattice by up to 26x, but also enabling vibration control via electrical shunting, shifting natural frequencies by ~2 Hz. The experimental results confirm that this platform can act as a self-sensing mechanical support suitable for structural health monitoring and adaptive vibration mitigation. The multidisciplinary concept across different applications is shown in Fig. 1.

3. Materials and methods

This research focuses on determining the feasibility of integrating smart materials into artificially designed mechanical metamaterial lattice structures through structural and coupled finite element analysis

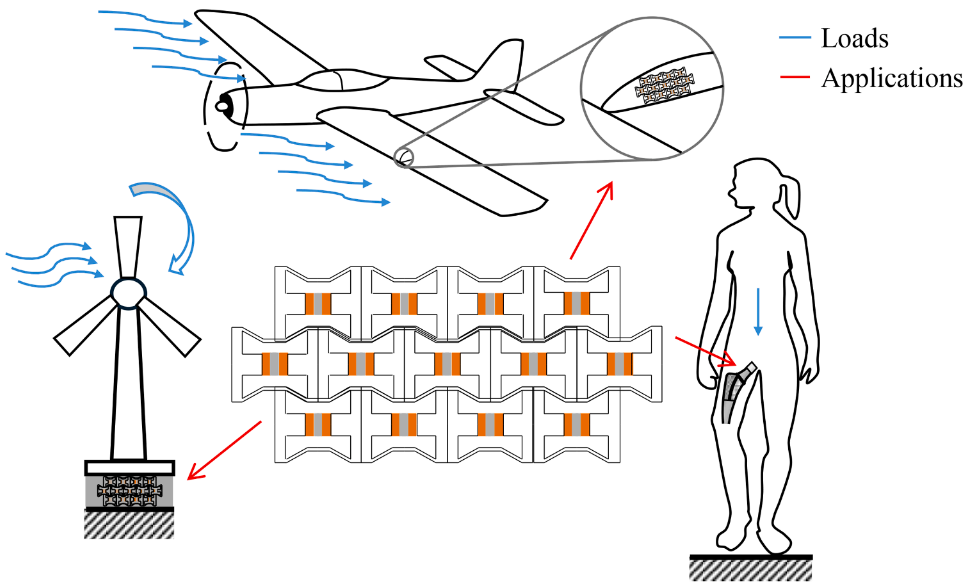


Fig. 1. Smart structure concept for a mechanical system under load, employing piezoceramic elements to improve the mechanical and sensing capabilities of the host structures.

(FEA). Another goal was to evaluate the performance and limitations of the sensing capabilities of smart structures fabricated via additive manufacturing that incorporate embedded piezoelectric sensors.

3.1. Design of multidisciplinary metamaterial structure

A bottom-up approach was selected to design a smart metamaterial lattice structure incorporating piezoelectric elements. The design process consisted of three steps, during which the smart structure was carefully evaluated using FEA and experimentally validated.

1. We designed an auxetic reentrant metamaterial unit cell with a tailored geometry for piezoelectric transducer integration and evaluated its response.
2. Scaling the base unit-cell geometry to a periodic lattice structure capable of bearing external elements and evaluating its performance, auxetic behaviour, sensitivity, and load-bearing capacity.
3. Design a smart-platform laboratory sample with added weight to mimic the sensing structure. This will evaluate the load-bearing capacity and sensing ability of such structures.

The foundation of our auxetic metamaterial lies in its reentrant hexagonal lattice structure, chosen for its inherent ability to exhibit a negative Poisson's ratio, a characteristic in which the material contracts laterally when compressed. Auxetic geometry was selected to exploit the unique mechanical properties of a negative Poisson's ratio, enabling it to deform uniformly under load while maintaining the overall structural shape. Fig. 2(a) illustrates a schematic of the unit cell, clearly depicting four interconnected struts forming T-shaped elements arranged in a honeycomb pattern. Each cell measured 22 mm in width and 26 mm in height at its outer vertices. The strut thickness was maintained at 0.5 mm to balance structural rigidity and facilitate integration of piezoelectric sensors within the unit cell's horizontal T-shaped ribs. The importance of strut thickness was also constrained from a manufacturing perspective, and a minimum strut thickness of 0.5 mm was chosen to

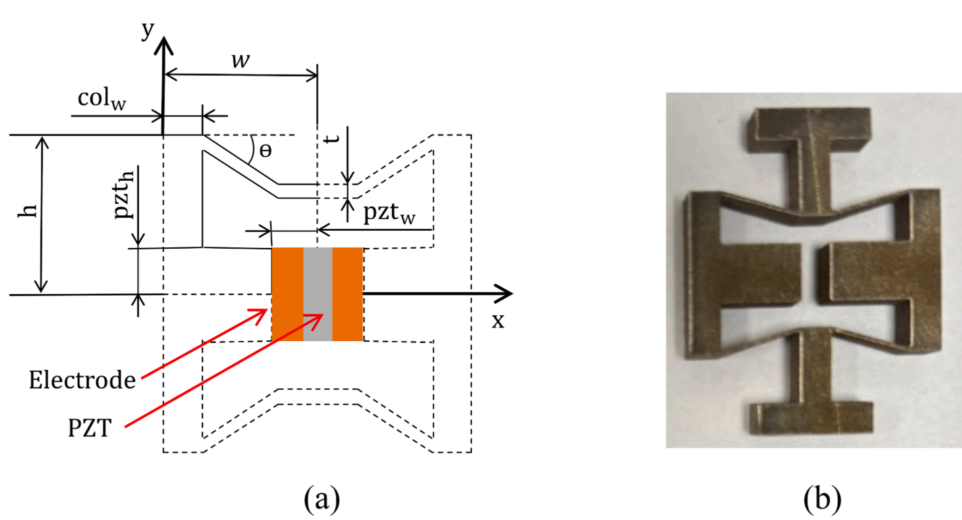


Fig. 2. Auxetic material characterisation. (a) Dimension schematic of a unit cell, where the x-y axis shows the coordinate system, then the dimension arrows describe the geometry with dimensions provided in Table 1 first row and red arrows marking the electrodes and PZT insertion (b) Photograph of the fabricated metallic auxetic component with T-shaped unit.

achieve desirable shape and dimensions and to assess the usability of additive manufacturing (Fig. 2(b)).

A key aspect of our design was to use T-shaped units and integrate piezoelectric sensing elements directly to enable in situ monitoring of the auxetic structure's loading. We opted for a piezoelectric ceramic element (PZT), NCE51 measuring 5 mm x 5 mm x 1 mm, strategically embedded within the horizontal T-shaped rib of each unit cell. Under compressive loading, stress is applied perpendicular to the loading axis in the PZT, thereby generating a sensing signal.

Fig. 2(a) highlights the precise placement of the PZT element at the intersection of the two struts, ensuring maximum strain transfer from the lattice structure to the sensor during deformation. The orientation of the PZT element was deliberately aligned with the primary loading direction to achieve optimal sensitivity to stress changes along that axis. Placement and alignment were crucial for maximising the sensor's effectiveness in capturing strain variations within the auxetic metamaterial while maintaining the structure's rigidity.

3.2. Wall model

Based on the unit-cell structure and periodic scaling (Fig. 3(a)) of the unit cell with integrated PZT modules, a 2D structure was created to further assess the usefulness of the set design. The dimensions of the model are presented in Table 1. The structure has three rows and 13 cells. The PZT modules were modelled in the middle row of the three unit-cells. This design stiffens the entire structure, thereby enabling distributed sensing throughout it. The wall model is shown in Fig. 3(b).

3.2.1. Static analysis

A series of computational models was used to evaluate the effects of the PZT modules on the structure. The load-bearing capacity of a stiffened structure with piezoelectric elements was first assessed using a 0.1 mm displacement-controlled compression test. A fixed reference point on the middle leg (Fig. 3(b)) prevents lateral deflection of the side stands under uniaxial loading and due to fabrication imperfections. Subsequently, an equivalent stress analysis of the auxetic wall was performed to evaluate the trusses' rigidity under a displacement load and to identify the structure's weak points. This led to a deformation analysis under a 0.5 mm displacement load with compression-only supports on the remaining legs. These conditions were employed to study an auxetic structure without PZT elements, with a focus on stiffening. A compression test was also performed on two additively manufactured Ti-6Al-4 V and SS316L structures, one with and one without integrated PZT modules. The structure without the PZT modules had no top or bottom stands, as in the static analysis, but used force-controlled cyclic loading rather than displacement control. A sinusoidal force with an amplitude of 1.4 N was applied across a frequency range of 100–1000 Hz.

Table 1
Base unit cell, 2D auxetic wall, and laboratory sample dimensions.

	w [mm]	h [mm]	col_w [mm]	$sens_h$ [mm]	$sens_w$ [mm]	Θ [deg]	t [mm]
Unit-cell	11	13	1.5	2.5	1.1	17	0.5
Lattice structure	79	37	1.5	2.5	1.1	17	0.5
Laboratory sample	95	97.5	1.5	2.5	1.1	17	0.5

3.2.2. Harmonic analysis

Harmonic analysis was performed on an auxetic wall equipped with PZT modules to investigate the potential to tune its natural frequency by varying the electrical load applied to the modules. All three PZT modules were connected in parallel to a resistive load. Two electrical loading conditions were considered: an open-circuit scenario, modelled with a high resistance of 10 MΩ, and a short-circuit scenario, represented by a low resistance of 100 Ω. The auxetic walls were subjected to the same loading configuration. Under open-circuit conditions, a peak displacement of 12.15 μm was observed at a loading frequency of 760.6 Hz. In the short-circuit case, a slightly higher peak displacement of 12.28 μm was observed at a lower frequency of 759.8 Hz. Results of numerical simulation of vertical displacement of the auxetic wall for boundary and resonance frequencies for the open-circuit configuration are shown in Fig. 4 (a-c), and for a short-circuit in Fig. 4(d-f). The impact of the change of the electric configuration on the mechanical behaviour of the auxetic wall is shown in Fig. 4(g), where amplitude of vertical displacement of the middle top point of the wall for both cases is compared. The relative change in resonance frequency normalised to open-circuit resonance frequency is 0.1 %.

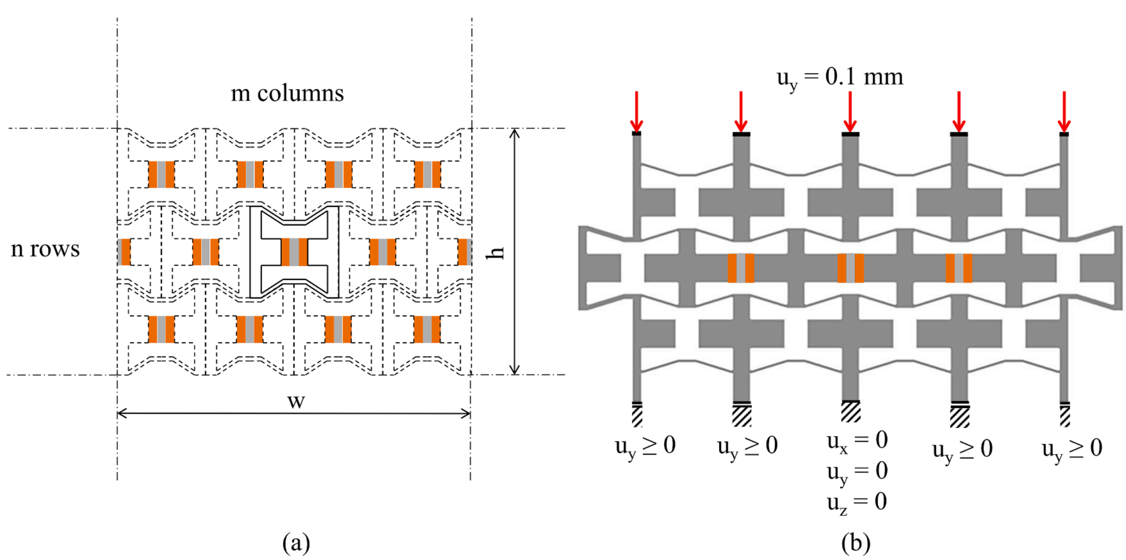


Fig. 3. Geometry model of auxetic wall for static analysis: (a) Unit cell in a periodic array, where m-n means the periodicity in both directions and w-h are the dimensions of our auxetic wall with dimensions in Table 1 second row. (b) Deformation boundary conditions for a static analysis, compression testing performed for stiffness comparison.

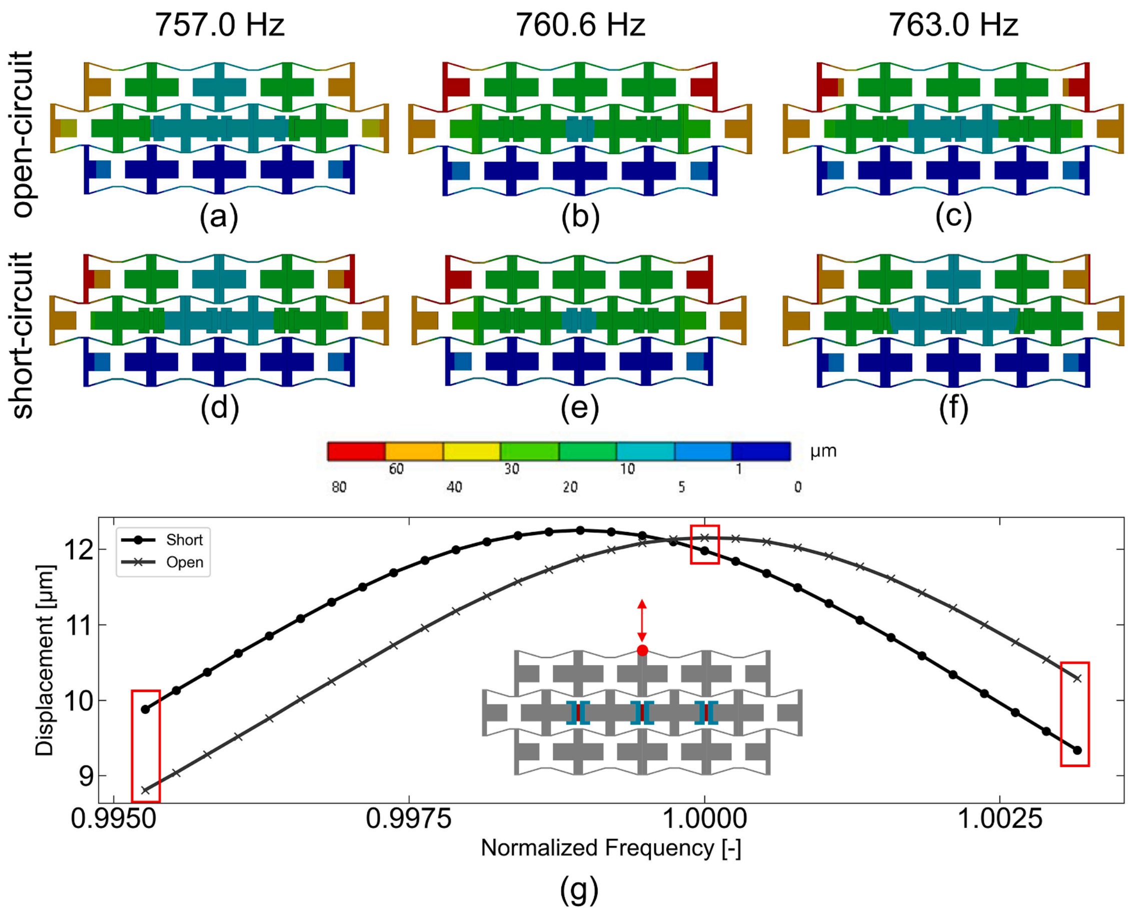


Fig. 4. Results of numerical simulation of vertical displacement of the auxetic wall over a frequency range from 757 Hz to 763 Hz with open-circuit and short-circuit PZTs. (a-c) open-circuit at 757.0 Hz, 760.6 Hz and 763.0 Hz, respectively; (d-f) short-circuit at 757.0 Hz, 760.6 Hz and 763.0 Hz, respectively; (g) graph of the vertical displacement of the middle top point of the wall over the frequency range normalized to the resonance frequency 760.6 Hz of the open-circuit scenario.

3.3. Computational modelling of multidisciplinary metamaterial

Computational modelling using the finite element method effectively and comprehensively assessed the behaviour of metamaterial structures with PZT. This enables the constraints, loads, and electromechanical properties to be modified relatively easily and efficiently, thereby affecting the observed output parameters. In this case, the effects of short- and open-circuit conditions on the PZT element's mechanical vibration modes were evaluated to assess the feasibility of using PZT modules at selected positions.

To test the functionality of our base unit cell design and confirm the effectiveness of the integrated piezoelectric sensor, we conducted finite-element method (FEM) simulations. Using ANSYS Academic Research Mechanical, 2024 R.2 (Swanson Analysis Systems Inc.), we modelled the deformation of the unit cell under various load conditions. This provides valuable insights into the distribution of stress and strain amplification at the sensor location. The FEA results clearly showed that our chosen placement and orientation of Lead Zirconate Titanate (PZT) effectively captured the maximum strain experienced by the auxetic structure during loading, demonstrating its suitability for enhancing stiffness.

Furthermore, coupled-field analysis was employed to perform static structural, modal, and harmonic analyses of piezoelectric materials. By applying a load to the top surface, having a correct coordinate system for the piezoelectric material, and appropriate connections and boundary conditions, an analysis can be performed to assess the chosen design and the effects of different parts of the structure on its behaviour. The auxetic structure uniformly deforms and responds to location and loading accordingly. The displacement in the T-shaped units strained the piezoelectric element, generating a voltage response. Structural deformation aids the interpretation of transient effects in unit cells for vibration sensing. Harmonic analysis is then used to assess vibration attenuation and determine whether a structure can control mechanical vibrations.

3.3.1. Geometry model and mesh

The geometry model was developed for an auxetic unit, an auxetic wall, and a smart platform. Further analysis was performed on the model of the experimental smart platform, as shown in Fig. 5(a). All relevant and significant geometric details are included in the geometric model. This model was discretised using structural solid elements SOLID186 and SOLID187, and coupled-field elements SOLID226 and CONTA174 for contact-target modelling (Fig. 5(b)). The model contains 934,207 nodes and 355,043 elements. The sensitivity analysis revealed that doubling the number of nodes and elements was unnecessary, as the difference in results between the finer and original meshes was less than 5 %. The piezoelectric elements require a coordinate system to specify the orientation of their polarisation. The coordinate system was rotated along the y-axis to target 33 modes in the piezoelectric matrix. When the structure is loaded, the appropriate coupled effects are applied to evaluate the model's electromechanical response.

3.3.2. Material model

The model was developed using the material library referenced in ANSYS2024, incorporating customised properties such as Young's modulus (E), Poisson's ratio (ν), and density (ρ). Various additive materials, including Ti-6Al-4 V, SS316L, and PLA, were modified to as closely as possible reflect the inherent nonhomogeneous state and reduced density resulting from inconsistencies in the additive

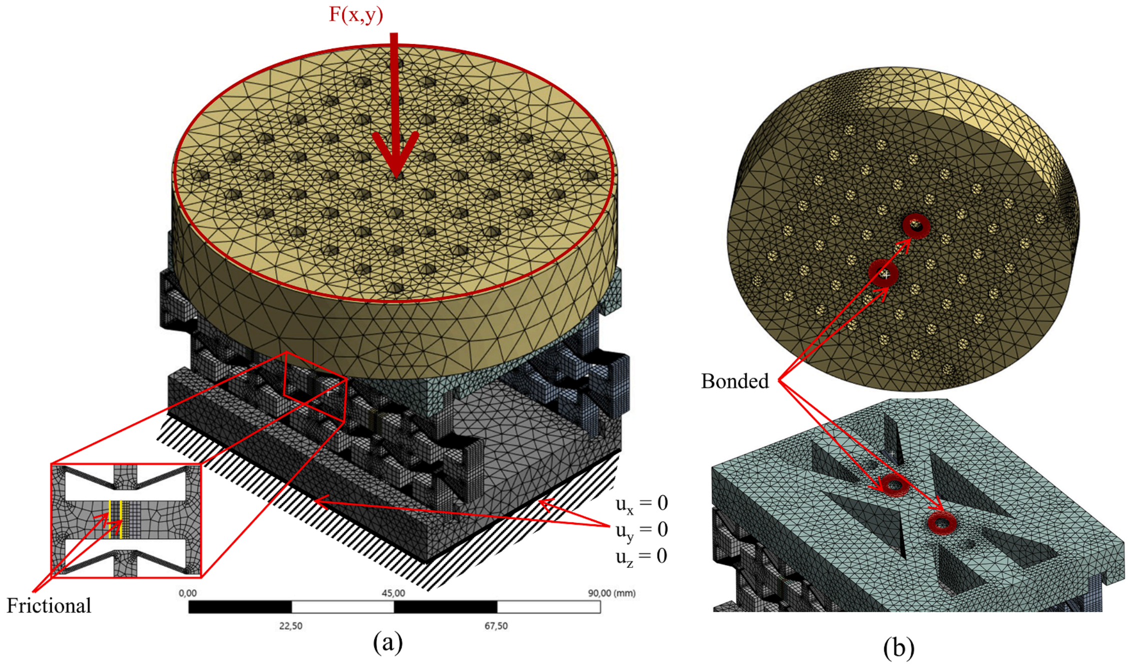


Fig. 5. Meshed model of laboratory sample: (a) global mesh showing load F(x,y), boundary conditions, and support; (b) detailed bolt contact region. The model contains 934,207 nodes and 355,043 elements (SOLID186/SOLID187 for structure, SOLID226/CONTA174 for piezoelectric contacts), with sensitivity analysis confirming <5 % difference when doubling mesh density.

manufacturing (AM) process. This was crucial for addressing the differences observed between the theoretical and actual fabricated models. In addition to these properties, another critical factor in refining the material model was comparing the computed weight of each element with that of a laboratory sample to ensure accurate representation. The PZT-5H model was chosen as the foundation for the piezoelectric material model, and several parameters were varied, including the piezoelectric matrix, anisotropic elasticity matrix, relative permittivity, and density (ρ). These adjustments aimed to align the material properties with those specified in the datasheet for NCE51 [76]. This piezoelectric material enables coupled-field analyses of static, modal, harmonic, and transient problems, yielding useful insights into complex structures incorporating piezoelectric materials. This must be reflected in the settings of the coupled-field analysis: in the Physics Region, select the structural and electric domains for specific elements in the geometry, and enable coupled-field effects. However, the material characteristics suitable solely for structural analysis are included in Table 2. Furthermore, the stock material FR-4 composite laminate is available in the General Materials Ansys library. In addition, stainless steel (SS) was incorporated, with the density adjusted to achieve a target weight of 1.06 kg. The material properties of the NCE51 piezoelectric material are listed in Table 3.

3.3.3. *Constrain and load*

The stainless steel and the top platform made of polylactic acid (PLA) were connected at the centre with two bolts. The connection in ANSYS must reflect this. Imprint faces and bond connections were employed to connect the weight and top platform to improve the modal simulation results (Fig. 5(b)) Because the PZT elements and spacers were in compression, a frictional connection with a coefficient of 0.4 was utilised. The specimen was secured on the underside with a fixed constraint and subjected to a remote force of 80 N at the centre (Fig. 5(a)). A solution for the coupled field static, modal, and harmonic analyses was obtained. Modal analysis was conducted for the six modes identified as the most dominant for the structure and added weight. Harmonic analysis was also performed to evaluate the mechanical vibration attenuation due to the electrical load around the most significant mode, which occurred around the peak mode, between 150 and 170 Hz.

Table 2
Material model characteristics.

	Young's modulus [GPa]			Poisson's ratio [-]			Density [kg/m^3]
	E_x	E_y	E_z	ν_{xy}	ν_{yz}	ν_{xz}	ρ
FR-4	20.4	18.4	15	0.11	0.09	0.14	1840
Ti-6Al-4 V [77]	87			0.323			4405
SS316L [78]	195			0.25			7954
PLA [79]	0.8			0.33			700

4. Design and model of smart platform

An auxetic wall with integrated PZT modules is a complex structure that facilitates the evaluation of fundamental structural stiffness and sensory capabilities. However, conducting experiments on such structures is challenging. The voltage response is insufficient for acquiring structural vibrations, as the output signal decays rapidly. This, coupled with the difficulty of using the wall as a laboratory platform, underscores the complexity of the need for a more resilient structure. It employs two auxetic wall samples, additively manufactured from SS316L. These two walls are parallel and separated by a 45 mm gap; they are mounted in the top and bottom fixtures, which are made of PLA. This changes the previous property of sensing impact along one dimension from the top of the auxetic wall to a two-dimensional sensing capability. If the structure is loaded at either the top or bottom fixture, the corresponding voltage levels of the PZT elements vary with the loading position. Each wall has three sensors located at the centre of three unit cells, covering the sensing surface. Owing to the negligible fixture mass of 30 g, loading the structure did not generate sufficient vibrations in the PZT elements for proper analysis. This necessitated the addition of a proof mass to simulate the structure's load-bearing capacity and to demonstrate the potential for intelligent support behaviour. With added mass, adequate testing is feasible because self-excitation and uniform damped oscillations are attainable. This also enables a more accurate assessment of vibrational performance at lower frequencies.

Table 3
Piezoceramic characteristics for NCE51.

	Anisotropic Elasticity x10^{12}[m^2/N]						Piezoelectric matrix x10^{-12} [C/N]		Density [kg/m^3]	Anisotropic Relative Permittivity [-]
	s_{11}^E	s_{12}^E	s_{13}^E	s_{33}^E	s_{55}^E	s_{66}^E	d_{31}	d_{33}	ρ	K_3^T
NCE51 [76]	17	−4.8	−8.9	21.3	49.0	43.4	−208	44	7850	1900

4.1. Modal analysis

Modal analysis focuses on identifying the structure's natural frequencies and mode shapes, which are critical for understanding its vibrational behaviour, particularly in the presence of added mass and piezoelectric components. Computational modelling enables evaluation of how weight, auxetic geometry, smart materials, and boundary conditions influence the dynamic response, thereby enabling optimisation in applications such as vibration control and structural health monitoring. Modal analysis was employed to compare with experimental modal analysis and to select the dominant mode for harmonic analysis or for vibration attenuation using PZT elements.

4.2. Harmonic analysis

Harmonic analysis of the smart platform was conducted using the same methodology as for the single auxetic wall. Six PZT modules (three embedded in each wall) were connected in parallel to a resistive load.

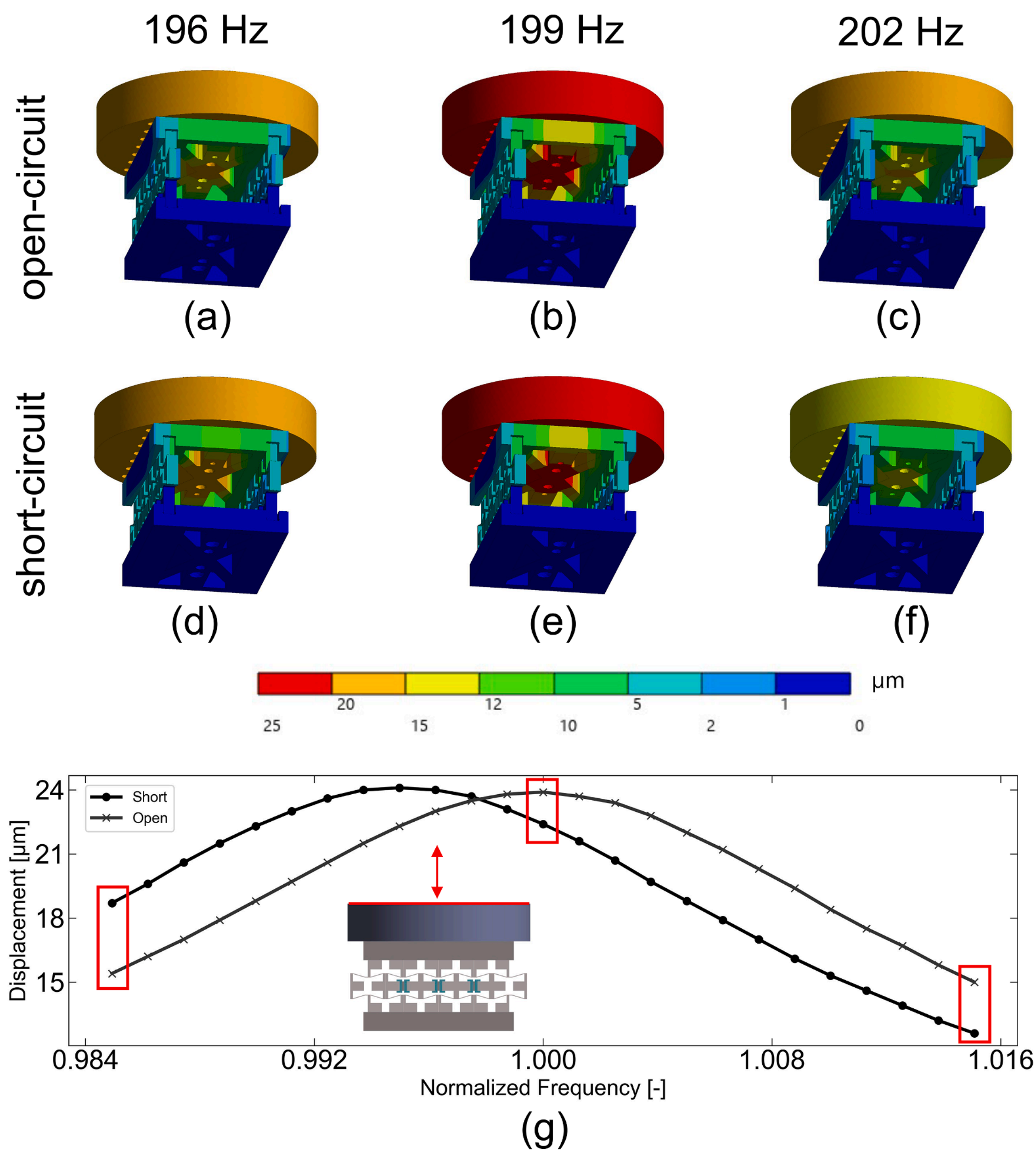


Fig. 6. Results of numerical simulation of vertical displacement of the smart platform over a frequency range from 196 Hz to 202 Hz with open-circuit and short-circuit PZTs. (a-c) open-circuit at 196 Hz, 199 Hz and 202 Hz, respectively; (d-f) short-circuit at 196 Hz, 199 Hz and 202 Hz, respectively, (g) graph of the average vertical displacement of the top face of the platform over the frequency range normalized to resonance frequency 199 Hz of the open-circuit scenario.

Two electrical loading conditions were examined: open-circuit and short-circuit, modelled with the same resistance values as in the previous analysis (10 MΩ and 100 Ω, respectively). A cyclic force with an amplitude of 1.4 N was applied over a frequency range from 196 Hz to 202 Hz. In the open-circuit configuration, the platform exhibited a peak vertical displacement of 23.9 μm at an excitation frequency of 199.0 Hz. In the short-circuit case, a peak displacement of 24.1 μm is observed at a lower excitation frequency of 198.0 Hz. Results of numerical simulation of vertical displacement of the smart platform for boundary and resonance frequencies for open-circuit configuration are shown in Fig. 6 (a-c), and for short-circuit in Fig. 6(d-f). The impact of the change in the electric configuration on the mechanical behaviour of the platform is shown in Fig. 6(g), where the vertical displacement amplitude at the platform top is compared for both cases. The relative change in resonance frequency normalized to open-circuit resonance frequency is 0.5 %.

5. Additive technology and manufacturing of auxetic structure specimens with PZT modules

The unit cell and auxetic wall were 3D-printed in PLA to assess the auxetic behaviour and sensing capabilities of the proposed design, which incorporates PZT sensing elements. Following tests of the uniform deformation resulting from the design, additive manufacturing was employed using Selective Laser Melting (SLM) with SLM 280HL (SLM Solutions Group AG, Lübeck, Germany) and fused deposition Modelling (FDM) 3D printing on a standard Prusa printer MK3S to create a unit cell and planar auxetic structure comprising three rows and five columns. The final laboratory unit cell and planar auxetic samples were fabricated from SS316L (Fig. 7(a)) and TI-6Al-4 V (Fig. 7(b)) using SLM. These samples were utilised to integrate the PZT module and examine the voltage response.

6. Experiment

Three test configurations were established according to the manufacturing process. First, 2 AM samples made from Ti-6Al-4 V with integrated PZT elements were used to evaluate the stiffness of the manufactured samples, both with and without PZT elements. Second, the auxetic 2D structure was equipped with a flex-PCB to attach PZT modules to the top and bottom housings for electromechanical response testing. Next, a platform with upper and lower fixtures was created to connect the two 2D auxetic structures that feature PZT elements, thus forming a sensing platform resembling the model in Fig. 5. To enhance the damped electromechanical oscillations of the PZT, a 1 kg stainless steel weight was added to transform an initially asymmetric response into a symmetric one, yielding a response with equal phase shift and uniform characteristics under a symmetric load.

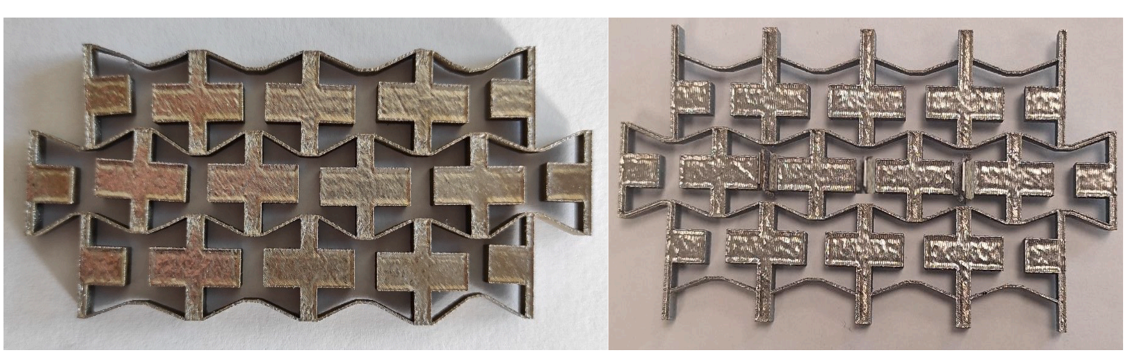
6.1. Experimental setup

Experimental tests were conducted to assess the feasibility of the proposed auxetic structures with embedded PZT elements under various loading conditions (random base excitation on a vibration shaker, impact hammer, and compression testing) across two laboratory configurations (an auxetic 2D structure and auxetic structures supporting a weight). To evaluate stiffness and changes induced by PZT elements, a compression test was performed on two 2D auxetic structures: one with and one without PZT elements. Two auxetic wall-structure samples were fabricated via selective laser melting (SLM) additive manufacturing and subjected to a compression test on a Step-Lab Corp. Static Material Testing Machine EA05 with displacement-controlled loading. The test sample consisted of three embedded elements with FR-4 spacers and electrodes, utilising PZT elements inserted under prestress to stiffen the core of the auxetic structure. A similar laboratory sample was fabricated using PZT elements from CTS Corp. of the hard material type NCE51, attached to a Flex-PCB, as shown in Fig. 8(a-b). The Flex-PCB features three pads for each positive-side electrode of the PZTs and a single common ground. The wires were connected to the NI-DAQ system, which was mounted with two NI cards (NI-9205 and NI-9234) featuring anti-aliasing filters and 24-bit analogue-to-digital (A/D) converters. The structure was excited with an impact hammer to evaluate responses at various impact locations on the impact surface and to observe the response signals of the embedded elements.

Following preliminary testing of the auxetic wall integrated with piezoelectric sensors, a laboratory platform was constructed to connect the upper and lower housings via rail slides, enabling distributed sensing along both axes. The final platform was firmly fixed to a testing anvil by using clamps attached to its lower housing. It was tested with and without an additional mass to assess its effect on the impact and output signal (Fig. 9(a-b)). Subsequently, impact tests were performed at nine distinct locations: six positioned on top of the PZT elements and three centrally aligned, parallel to the six locations mentioned above. The tests elicited an electromechanical response and excited six vibrational modes inherent to the structure. This experimental procedure yielded significant insights into location-dependent effects and the amplitude of the applied force. Additionally, it presented the potential to utilize the PZT output to extract information about the vibrational modes of the structure, thus facilitating structural monitoring applications.

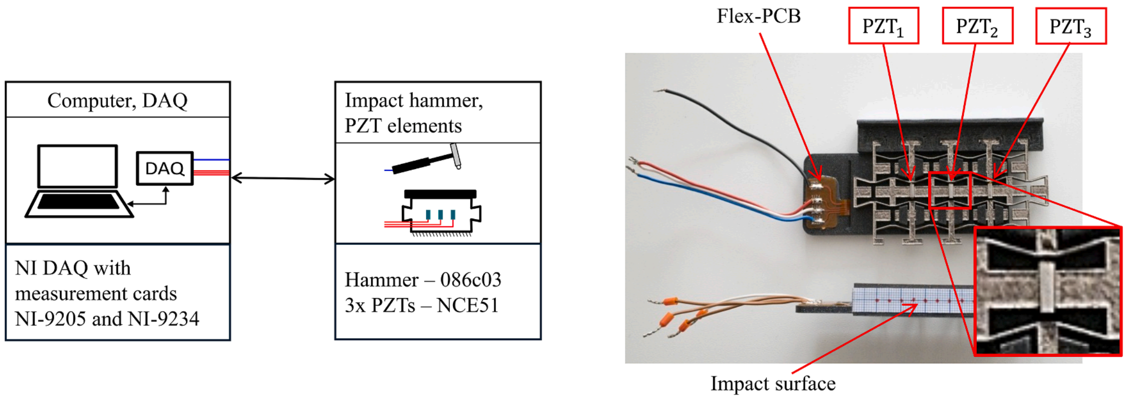
Following this, a vibrational test employing an electrodynamic shaker was conducted; specifically, a sweep test around the dominant natural frequency was performed with three distinct test configurations to evaluate the feasibility of using embedded PZT modules for vibration control. The structure was subjected to a sweep test in the frequency range 150–170 Hz. The sweep test lasted four minutes. It was performed with a constant acceleration amplitude of 0.05 g.

The PZT modules were available in three configurations: open-circuit, 10 Ω, and 560 Ω. The mechanical response of the structure is



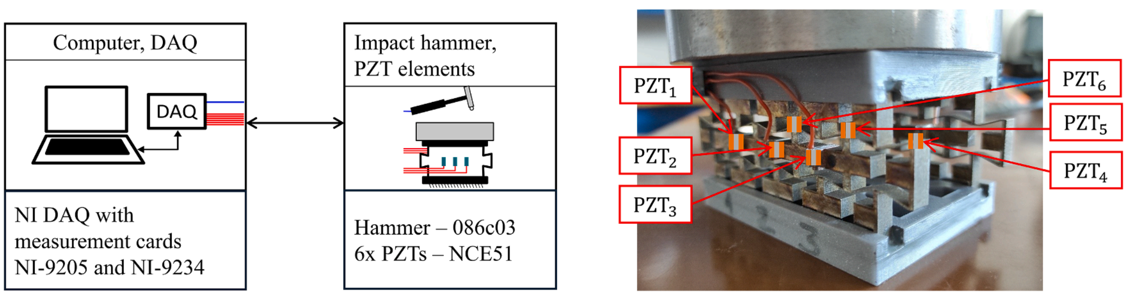
(a) (b)

Fig. 7. Additively manufactured auxetic walls. (a) from SS316L material used in the platform, (b) Sample of an auxetic wall and unit cell made from Ti-6Al-4 V.



(a) (b)

Fig. 8. Experimental impact testing setup for an auxetic wall with embedded PZT elements. (a) Schematic showing the data acquisition system (NI-9205 and NI-9234 cards with anti-aliasing filters, connected to a computer via DAQ), impact hammer, and three PZT elements (NCE51 type) a) diagram of the experimental setup, (b) Photograph of the fabricated auxetic wall sample with three embedded PZTs (PZT_1–PZT_3), FR-4 spacers, and Flex-PCB electrode pads. The structure was tested for initial sensing results and for short- and open-circuit electrical measurements.



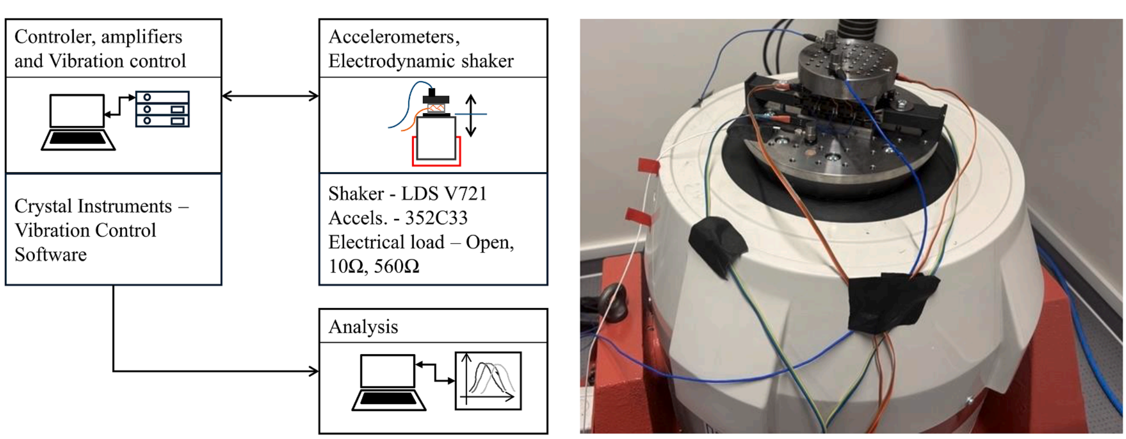
(a) (b)

Fig. 9. Experimental impact testing setup for a smart platform with embedded PZT elements. (a) Schematic showing the data acquisition system (NI-9205 and NI-9234 cards with anti-aliasing filters, connected to a computer via DAQ), impact hammer (086c03), and six PZT elements (NCE51 type), (b) Photograph of the fabricated smart platform with six embedded PZTs (PZT_1–PZT_6), rail slides connecting upper and lower housings, and electrode pads. The structure was tested with and without additional mass to assess the effect of mass addition on impact response, with impact tests conducted at nine distinct locations.

as follows: measured with two accelerometers at perpendicular locations on the edge of the added mass, as indicated in Fig. 10.

6.2. Stiffness of auxetic wall

The stiffness of the structures plays a crucial role in the load-bearing



(a) (b)

Fig. 10. Mechanical resonance control setup with PZT modules and accelerometers. (a) Schematic showing the vibration control system with controller, amplifiers (Crystal Instruments), electrodynamic shaker (LDS V721), two accelerometers (352C33), and PZT modules configured in open-circuit, 10Ω, or 560Ω electrical loads, (b) Photograph of the platform mounted on the electrodynamic shaker with accelerometers attached to the edge of an added mass at perpendicular locations. The setup was used to analyse mechanical resonance responses under different electrical load conditions.

performance. A compression test for a 0.1 mm displacement load was used to determine the stiffness of the auxetic structure with and without PZT elements (Fig. 7(a-b)). It shows significant stiffening of the wall thanks to an embedded PZT element. Corresponding computational results were performed during static analysis (Section 3.3.1). The results show the directional displacement and equivalent stress distribution for the auxetic wall for the configuration with (Fig. 11(a-b)) and without (Fig. 11(c-d)) PZT modules. The comparison of the two configurations from the experimental compression test and the static analysis shows a substantial increase in stiffness between them. The stiffness of the specimen was measured, and the measurements were normalised by its density, yielding approximately 28 and 10 stiffness increases for the experiment and simulation, respectively. This is primarily due to differences in fabrication, boundary conditions, the presence or absence of PZT in the wall as the precision of their mounting to the lattice, inaccurate material parameters, and the overall porosity of the SLM samples (Fig. 7). Testing the porosity of the SLM material has been not tested as for example authors in [80] proved that the Young's modulus depends also on the wall thickness. However, the goal of this paper was the proof-of-concept revealing the possible critical spots arising from manufacturing process and computational modelling.

6.3. Sensing results

The sensing results from the impact testing of the smart platform indicate that the structure without added mass exhibits rapid decay (Fig. 13(a-e)). The vibrations within the structure dissipated quickly, with a noticeable decline observed shortly after the initial impact. This behaviour suggests a strong dampening effect when only the PLA fixture was employed for auxetic walls without added weight. Moreover, adding mass atop the fixture significantly affects the output signals, providing crucial information regarding the system's performance. These results highlight the response characteristics of the structure under dynamic loading conditions (Fig. 13(f-j)). A uniform electrical response was generated by loading the system at the centre of the added mass. The structural test points were located directly above the mass. The charts display the averaged responses of PZT (1, 3, 4, 6) at each location, as well as the average loading force. It also shows shaded

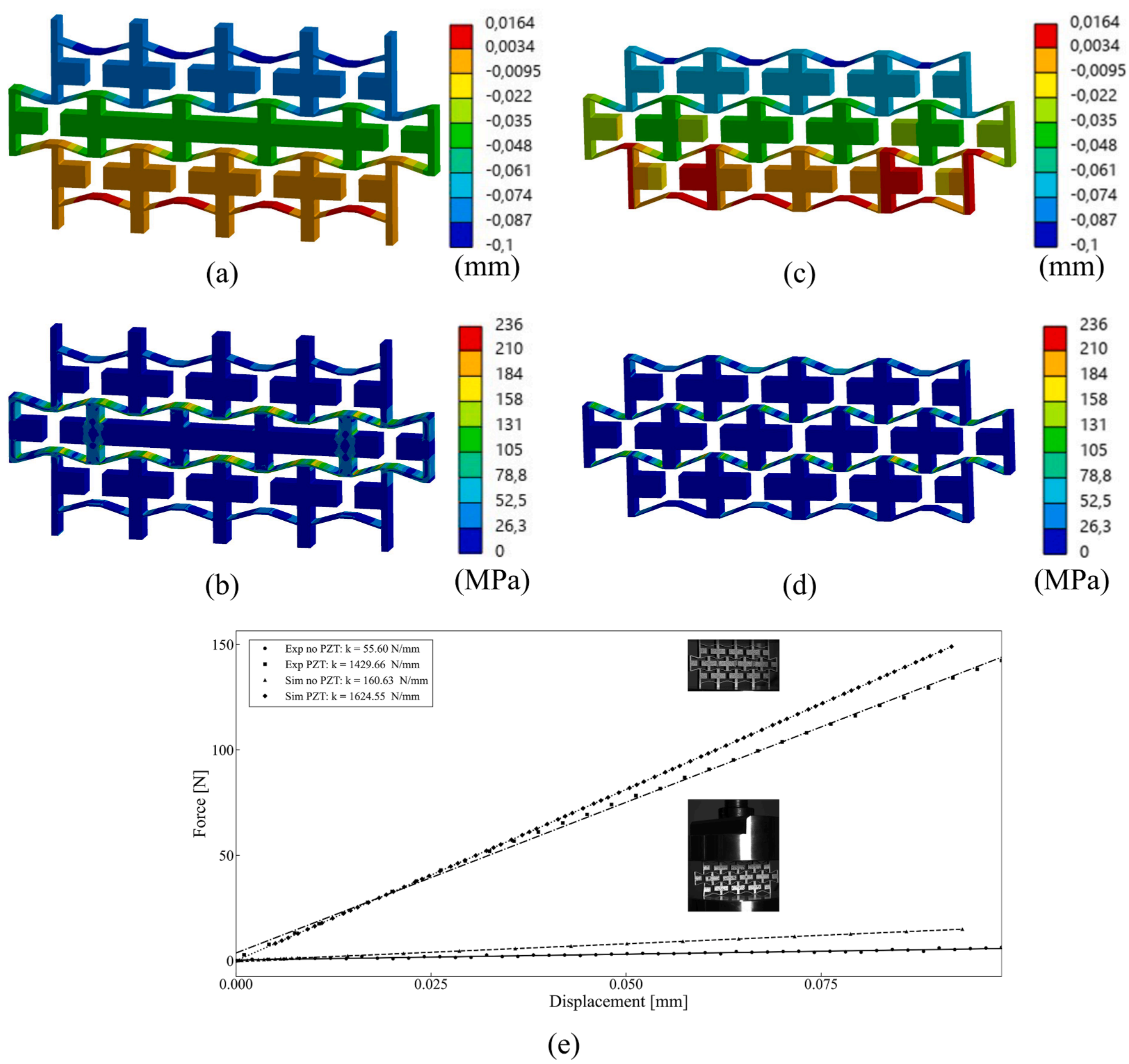


Fig. 11. Directional deformation, stress distribution and stiffness comparison of the auxetic wall. (a-b) Simulation results for the static analysis described in Section 3.2.1 depicting deformation for the auxetic wall with PZT in the loading direction and the von-Mises equivalent stress distribution, respectively. (c-d) Simulation results for the static analysis described in Section 3.2.1 depicting deformation for the auxetic wall without PZT in the loading direction and the von-Mises equivalent stress distribution, respectively. (e) Force-displacement measurements of auxetic reentrant lattice walls, comparing experimental (circles/squares) and simulated (triangles/diamonds) data for configurations with and without piezoelectric transducers (PZTs). Both sets show a significant increase in stiffness when PZTs are incorporated (stiffness, k, rises from ~55.60 to ~1429.66 N/mm experimentally and from ~160.63 to ~1624.55 N/mm in simulation), with the simulated trends closely matching the experimental measurements.

individual measured signals. The structural test points were located directly above the PZT modules and between the opposite points (Fig. 9 (b)). Location 0 marks the centre of the weight, with locations 1–6 corresponding to the positions above the numbered PZT modules. The electrical response to mechanical excitation exhibits lower amplitudes for sensors located farther from the excitation point, with appropriate phase shifts between the sensors' responses. The impact of the added mass is generally shifted relative to the electrical response, in contrast to the structure without an added mass, indicating a lag in the structural response of the PZT units.

In addition to the averages in the Fig. 13(a-j), all the individual measurements for each PZT are shown as semi-transparent traces, demonstrating repeatability of the sensing behaviour. Although the time-of-arrival differences are not explicitly highlighted, they are clearly visible in the waveforms: for most impact locations, the PZT closest to the excitation consistently shows the highest amplitude and the earliest response, demonstrating the platform's inherent localisation potential. The only notable deviation occurs at location 1 (Fig. 13(g)) above PZT1, when the impact on the mass is applied, the signal quality for PZT1 deteriorates relative to other locations. It is most likely due to damage or cracking in the PLA support structure caused by repeated testing. In the central region (Fig. 13(f)), the responses across sensors remain highly similar, as expected under symmetric excitation. Minor timing variations arise from the manual nature of impacts. Overall, the results support the conclusion that the sensing platform provides repeatable measurements and maintains spatial sensitivity across most tested locations.

Numerical and experimental modal analyses, as well as PZT signal modal analysis, were performed to verify the expected natural frequencies from the numerical model and to validate the usability of the PZT modules for harmonic analysis of the structure and the effectiveness of the embedded piezoceramic actuator in its implementation. The third mode was not visible during the measurements. The numerical modal analysis indicated that mode 3 (Table 4) has a lateral expansion (Fig. 12 mode 3 – 105 Hz) of the weight that could not be measured by PZT or uniaxial accelerometers. The large mass of the added weight and high energy within the dominant resonant mode 5 in Fig. 12 possibly leading to low-energy modes being buried in the noise.

6.4. *Influence of electrical load on mechanical resonance*

The frequency response analysis of this structure indicates a dominant mode near 155 Hz. Consequently, this mode was selected to evaluate the influence of the electrical load on the PZT elements and overall mechanical response of the structure. The vibrational testing sweep-up and sweep-down characteristics revealed frequency shifts of 2 Hz and 1.8 Hz under a 10 Ω load, respectively. The amplitudes at 155.4 Hz with a 10 Ω load are over twice as low as those observed during open-circuit sweep-up, and three-fifths of the amplitude recorded at 155.1 Hz during sweep-down. This property presents an opportunity to stiffen the structure and control its mechanical vibrations via electrical regulation (Fig. 14). Controlling the unit cell's stiffness within the periodic wall under electrical load shifts the natural frequency and significantly reduces the resonance amplitude, potentially mitigating failure states.

Table 4
Calculated and measured natural frequencies.

	Mode 1 [Hz]	Mode 2 [Hz]	Mode 3 [Hz]	Mode 4 [Hz]	Mode 5 [Hz]	Mode 6 [Hz]
Numerical	72	99	105	131	163	208
PZT response	79	85	-	138	164	220
Experimental	75	84	-	122	151	210

7. Discussion

This study explores the development of a multidisciplinary mechanical metamaterial platform featuring integrated piezoelectric ceramic sensors, which may represent a promising paradigm shift in embedded sensors and smart structures. The piezoelectric ceramic material within the structure employs auxetic structures for load-bearing supports and sensing, without compromising structural stiffness. The negative Poisson's ratio and reentrant design give the structure a lightweight property, as one of the main features of auxetic metamaterials [62]. However, the integrated PZT elements within the reentrant unit cells enable multiple innovative applications of this property. Firstly, the incorporation of three PZT modules into the auxetic wall provides the structure with almost 26 times greater stiffness, directly enhancing its load-bearing capacity (Fig. 11(e)). This effect aligns with previous research on auxetic structures and their ability to improve mechanical properties through geometric design or the innovative use of negative Poisson's ratio [33,68,72].

The vibration control potential was tested on a multidisciplinary model of an auxetic wall and PZT modules, utilising both short-circuit and open-circuit conditions. The frequency range selected for this test was the vertical acceleration mode, from 757 Hz to 763 Hz. The vertical resonance mode was shifted by approximately 1 Hz, providing a test of potential attenuation of mechanical vibration. Following this, a multidisciplinary model of the entire platform, encompassing static, modal, and harmonic analyses, was developed. The frequency range around the strongest vertical-acceleration mode was selected to examine the effects of vibration attenuation on the platform's mechanical resonance and the electrical load, spanning 196–202 Hz. This also showed a shift of around 1 Hz of the vertical mechanical resonance mode. The creation of multidisciplinary models provided a basis for fabrication, validation, and experimental testing.

The experimental results from impact loading on different locations of the platform (Fig. 13(a-j)) provide insight into the importance of added weight on the sensing performance of auxetic walls with the PZT elements. When the auxetic walls were not used as support, they exhibited rapid signal decay. However, with the added weight and thus the utilisation of the support potential, the electrical response became uniform. There is also a noticeable difference in the signal shift and amplitude depending on the loading location and the proximity of the PZT element. This also suggests the usability in the impact location and possibly the load reconstruction potential.

One of the most important findings is the effect of electrical loading on the mechanical vibration response. The effects of vibration reduction were tested on the strongest mode, which had a vertical mode shape in the range of 150–170 Hz (Fig. 14(a-b)). Under open-circuit conditions, the platform exhibited peak accelerations of approximately 21.5 m/s^2 at 155.4 Hz during upward sweeps and 21.2 m/s^2 at 155.1 Hz during downward sweeps. When a 10 Ω electrical load was applied, the acceleration was roughly 9 m/s^2, and 12.1 m/s^2 was observed at the same frequencies. The peak acceleration shifted to 157.4 Hz during the sweep-up, corresponding to a 2 Hz increase in the mechanical resonance frequency. When sweeping downward, the acceleration peak occurred at 156.8 Hz, approximately 2 Hz lower than the open-circuit peak. Previous research [11–13] has also noted similar effects of electrical load changes on shifts in mechanical resonance, which affect structural damping. This platform, with added weight, indicates that mechanical vibrations can be controlled and reduced even in more complex structures.

Several limitations must be addressed before practical implementation. The brittle nature of ceramic PZT imposes constraints on fatigue life under cyclic loading, necessitating future investigation into alternative composite piezoelectric materials that balance high electromechanical coupling with improved toughness [81]. Additive manufacturing tolerances also affect lattice geometry, influencing both stiffness and sensor performance; topology optimisation techniques with

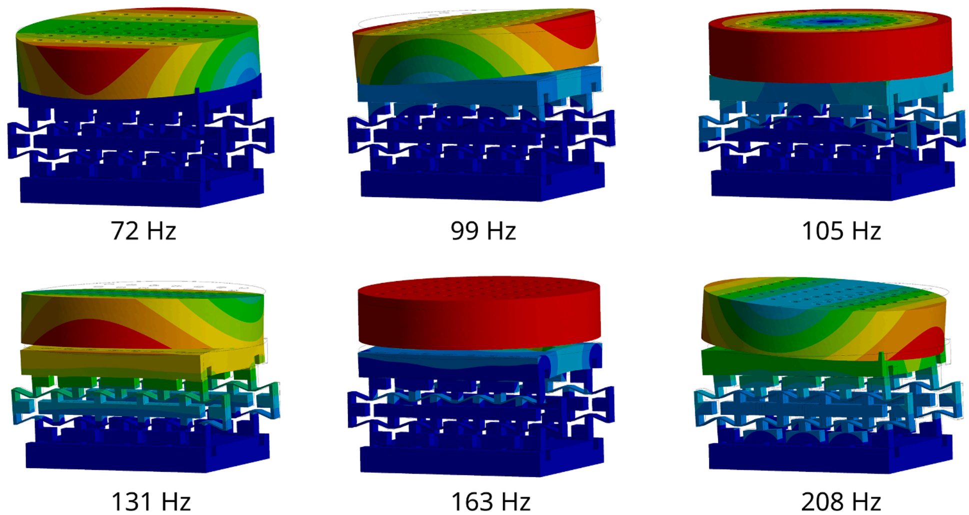


Fig. 12. Simulation results for modal analysis of a smart platform depicting all 6 modes together with the dominant fifth mode given by the added weight and used for harmonic analysis and vibration control. The first row shows mode 1–3, and the second row shows mode 4–6.

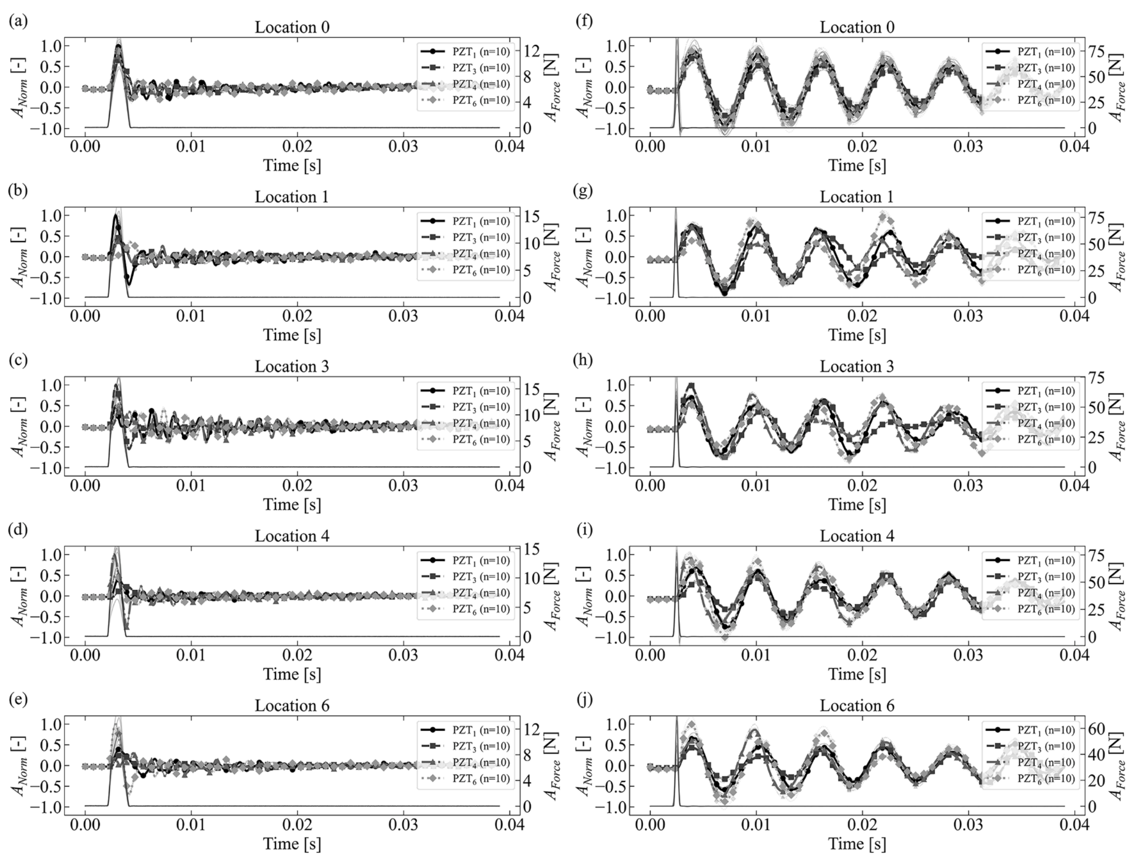


Fig. 13. Electrical response of the smart platform to impact loading. (a–e) Responses for the configuration without added mass; (f–j) responses with an added mass. Each subplot corresponds to one of six impact locations (0 = centre; 1, 3, 4, 6 = four corners). Signals from four piezoelectric transducers (PZT 1–4) and the force impacts are shown as bold lines (means of ten trials); the individual measurements appear as thin grey lines together with the corresponding force trace.

embedded sensors could enhance robustness to process variability while optimising sensing performance. Furthermore, the current implementation relies on external resistive shunts, which could be replaced by active shunting electronics (e.g., voltage-controlled oscillators or adaptive filters) integrated within the lattice structure. This advancement would enable real-time vibration attenuation and unlock energy-harvesting potential for autonomous sensing without compromising structural integrity [12,82].

Additionally, optimal placement methodology for piezoelectric materials in mechanical structures remains a challenge, requiring careful consideration during fabrication to ensure reliability [83,84]. Future work should focus on exploring different piezoelectric material configurations, with emphasis on fabrication techniques, signal interpretation, geometric design, and their interplay with structural performance. Addressing these challenges is essential to transitioning this proof of concept into practical smart structures for real-world applications.

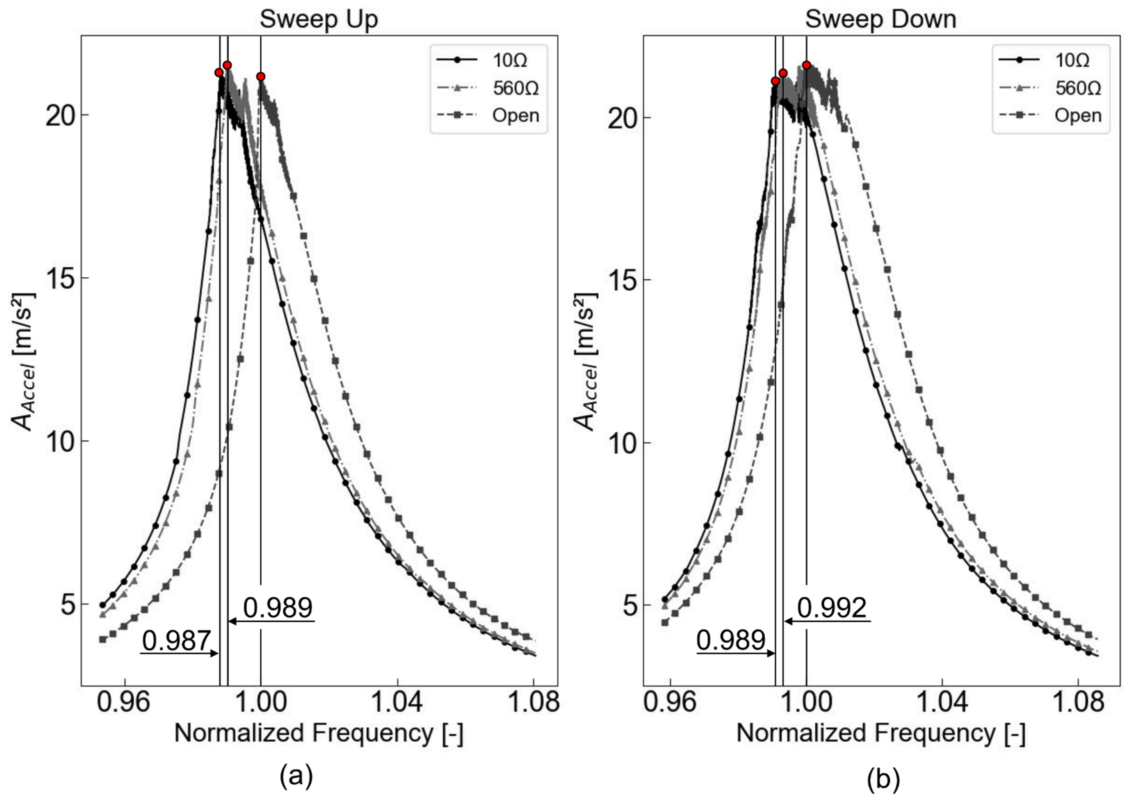


Fig. 14. Effect of electrical load on dominant natural frequency of fifth mode around a value of 160 Hz (a) Sine sweep up the frequency testing for Open-circuit and an electrical load of 10 Ω and 560 Ω, testing how an electrical connection on the PZT elements influences the mechanical behaviour of the platform, b) Sine sweep down around the dominant mode to see the effect of electrical load on the different direction of loading and its appropriate electrical load such as Open-circuit and an electrical load of 10 Ω and 560 Ω.

8. Conclusion

This study demonstrates a novel approach: embedding piezoelectric ceramic (PZT) elements within a reentrant auxetic lattice fabricated from Ti-6Al-4 V and SS316L by selective laser melting. This structure serves as both a lightweight mechanical support and a self-sensing unit capable of detecting external loads and controlling its dynamic response through electrical shunting. Computational modelling results showed that the PZT transducers would increase the axial stiffness of the wall by more than an order of magnitude (∼26x relative to unreinforced lattice) and introduce measurable electromechanical coupling. Experimental measurements support numerical simulations: compression tests showed a similar degree of stiffness amplification. In contrast, harmonic analysis showed a ∼2 Hz shift in the dominant natural frequency when switching the electrical load from an open-circuit (10 MΩ) to a short-circuit (100 Ω). These results indicate that simple resistive shunting can tune the structure's dynamic response.

Impact tests on the platform using an impact hammer with a vinyl cap further illustrate the smart platform's dynamic sensitivity. As shown in Fig. 13(a-e) and (f-j), the structural response is distinct depending on the loading configuration. Without added weight Fig. 13(a-e), the platform exhibits a stiff transient response, in which the signal energy dissipates rapidly within 0.01 s. In contrast, adding a ∼1 Kg weight atop Fig. 13(f-j) excites the dominant frequency primarily, as specified in the structure, and exhibits amplitude and phase variations with load location across the PZT sensors. This supports the practical viability of this approach for predicting potential load locations or fault localisation based on these signals. Moreover, modal frequencies obtained from both experimental modal analysis and PZT-based impedance measurements matched closely with numerical predictions (Table 4), supporting the accuracy of the computational coupled-field model. These findings open new possibilities for structural health monitoring without external instrumentation.

In summary, this study provides a proof of concept for a reentrant auxetic metamaterial platform that simultaneously functions as a load-bearing support and a self-sensing unit. This paper continues the development of a self-sensing and self-actuating smart structure with embedded PZT elements within a reentrant mechanical metamaterial structure, taking advantage of force amplification geometry and low frequency adaptation of piezoelectrics and following the idea of an integrated smart structure with the abilities of tunable dynamic response, stiffness amplification, impact sensitivity and additive manufacturing use case for complex geometries [13,74,75]. The demonstrated stiffness improvement, electrically controllable natural mechanical frequency, and clear impact-induced voltage signals offer a new perspective on smart structural components for civil, aerospace, and biomedical applications. Future research should focus on material durability, large-scale sensor integration, active vibration control circuits, and implementation in real-world loading scenarios, such as support for smart bearings, in-place functional replacement of existing elements within the structure, and studies on the use of bone hip implants.

CRediT authorship contribution statement

Vojtěch Slabý: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jan Bajer:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Petr Marcián:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Miroslav Hrstka:** Investigation, Formal analysis, Conceptualization. **Zdeněk Hadaš:** Writing – review &

editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data supporting this study's findings are openly available in the Zenodo repository at https://doi.org/10.5281/zenodo.15301752.

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